TOPIC 2: OPERATING SYSTEMS

WINDOWS AND LINUX FUNDAMENTALS FOR CYBERSECURITY

What is an Operating System?

- An Operating System (OS) is system software that manages hardware resources and provides common services for computer programs.

- It acts as an intermediary between the user, applications, and the physical hardware.

- Key functions of an OS include:

  + Process management: Creating, scheduling, and terminating running programs.

  + Memory management: Allocating and protecting Random Access Memory (RAM).

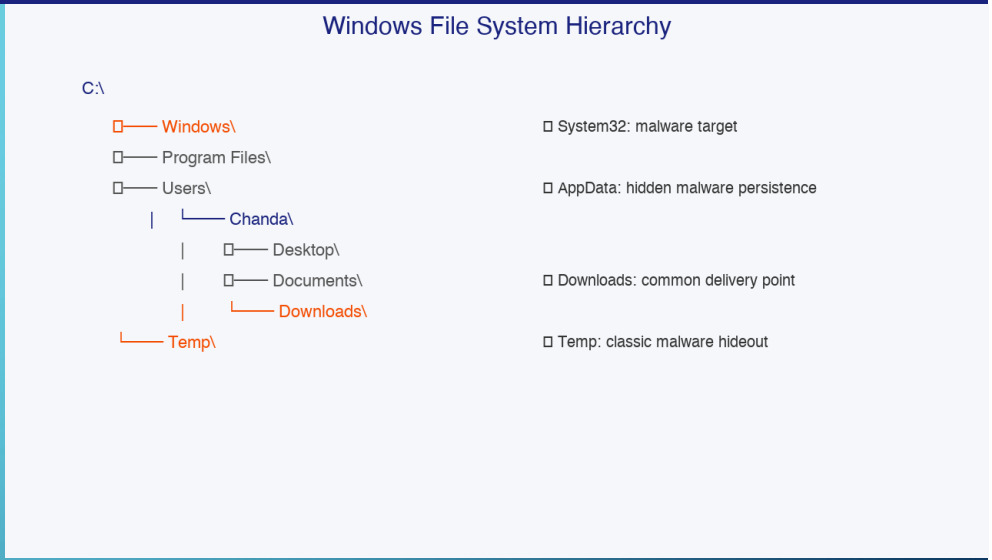
  + File system management: Organizing, reading, and writing files on storage devices.

  + Security and access control: Managing user accounts, passwords, and permissions.

  + Device management: Communicating with hardware through drivers.

Windows Operating System: File System Basics

- Windows is the most widely used desktop operating system globally and in Zambia.
- The file system is organized in a tree structure starting from drive letters like C:\.
- **C:\Windows\System32** contains core operating system files, Dynamic Link Libraries (DLLs), and executables. This is a frequent target for malware.
- C:\Users\[Username]\AppData is a hidden directory that stores application data. Malware frequently uses this location for persistence because it is writable by standard users.
- C:\Temp and C:\Windows\Temp store temporary files. Incident responders check these directories for recently dropped malware.
- C:\Program Files and C:\Program Files (x86) contain installed applications. Unauthorized software here may indicate a security breach.



Windows Command Prompt: Essential Security Commands

- ipconfig: Displays Internet Protocol (IP) configuration, including the computer's IP address, subnet mask, and default gateway. Using ipconfig /all shows additional details including the Media Access Control (MAC) address and Domain Name System (DNS) servers.

- netstat -an: Displays all active network connections and listening ports. Analysts use this to spot suspicious connections to unknown remote addresses.

- tasklist: Lists all running processes. Using tasklist /svc shows which Windows services each process is hosting, helping identify injected or rogue processes.

- systeminfo: Displays detailed system information including the operating system version, installed hotfixes (security patches), and hardware details.

- net user: Lists local user accounts. Attackers often create hidden backdoor accounts with innocent-sounding names.

- dir /a: Lists all files in a directory, including hidden and system files.

Windows User Accounts and the Principle of Least Privilege

- Administrator Account: Has full control over the system. Can install software, modify all files, and change security settings. Should be used only when necessary.
- Standard User Account: Has limited privileges. Cannot install software system-wide or modify protected operating system files. This is the recommended account type for daily use.
- Guest Account: Has very limited access and cannot make persistent changes to the system.
- The Principle of Least Privilege means giving users only the minimum level of access they need to perform their job.
- Security impact: If a standard user account is compromised by malware, the malware's ability to damage the system is limited. If an administrator account is compromised, the attacker gains full control.

Linux Operating System: Why It Matters in Cybersecurity

- Linux is an open-source operating system based on Unix. The source code is freely available, and anyone can inspect, modify, and distribute it.

- Linux powers the majority of web servers, cloud infrastructure, and supercomputers worldwide. Over 80 percent of web servers run Linux.

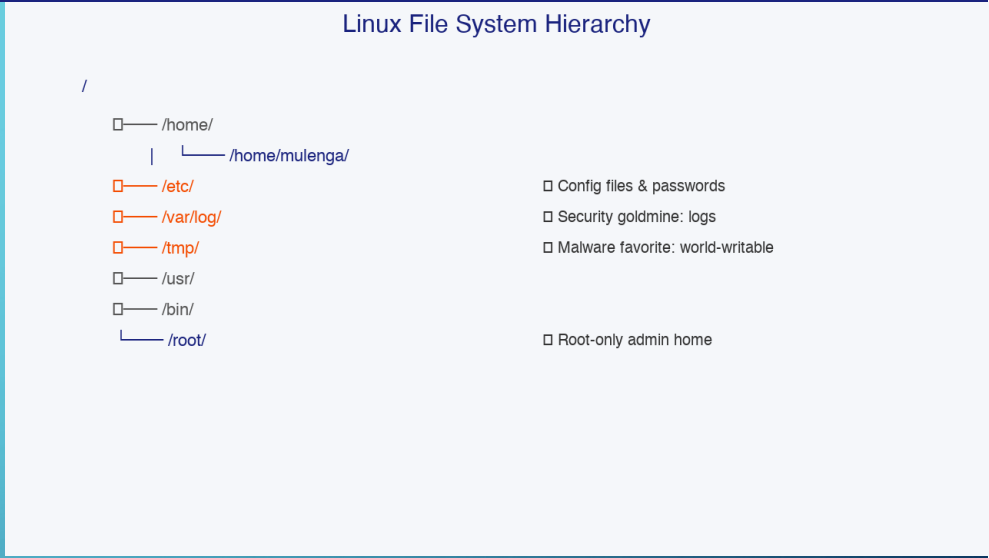
- In Zambia, core infrastructure at banks, mobile network operators, and government institutions runs on Linux servers.

- Kali Linux is a specialized Linux distribution designed for cybersecurity professionals. It comes pre-installed with hundreds of penetration testing and forensics tools.

- Because Linux is free, highly configurable, and stable, it is the platform of choice for most security tools and research.

Linux File System Hierarchy

- The Linux file system is organized as a single tree starting from the root directory, represented by a forward slash (/).
- /bin/ contains essential system commands available to all users, such as ls (list files), cat (display file contents), and chmod (change file permissions).
- /etc/ contains system-wide configuration files. Critical security files here include /etc/passwd (user account information), /etc/shadow (encrypted password hashes), and /etc/hosts (local hostname mappings).
- /home/ contains personal directories for each user. For example, /home/mulenga/ belongs to the user named mulenga.
- /var/log/ contains system log files. This is where authentication attempts, system errors, and application activity are recorded.
- /tmp/ is a temporary directory that any user can write to. Malware frequently drops files here.
- /root/ is the home directory of the root user - the Linux equivalent of the Windows Administrator account.



Essential Linux Commands for Security Professionals

- ls -la: Lists files in long format, showing permissions, owner, size, and modification date. The -a flag includes hidden files (files starting with a dot).

- cat /etc/passwd: Displays the contents of the /etc/passwd file, which lists all user accounts on the system.

- grep 'Failed' /var/log/auth.log: Searches the authentication log for lines containing the word 'Failed.' This helps identify brute-force login attempts.

- ps aux: Displays all running processes on the system, including those started by other users and system services.

- sudo chmod 600 file.txt: Changes file permissions so that only the owner can read and write the file. The group and others have no permissions.

- whoami: Displays the username of the currently logged-in user.

- history: Shows the command history for the current session. Attackers may delete or modify this to hide their tracks.

Understanding Linux File Permissions

- Every file and directory in Linux has three sets of permissions: one for the owner, one for the group, and one for all other users.
- The three permission types are: Read (r), Write (w), and Execute (x).
- Read (r) allows viewing the contents of a file or listing the contents of a directory.
- Write (w) allows modifying a file or adding and removing files from a directory.
- Execute (x) allows running a file as a program or script. For directories, it allows entering the directory.
- Permissions can be represented numerically: Read = 4, Write = 2, Execute = 1. The values are added together for each category.
- Example: chmod 755 script.sh means the owner has read, write, and execute (7 = 4+2+1), while the group and others have read and execute only (5 = 4+0+1).

Linux File Permissions

-rwx-xr-- 1 mulenga staff 1024 Mar 2028 script.sh

File Owner (rwx) | Group=(r-x) (r--)

Permission	Symbol	Value	Example
Read	r	4	• View file contents
Write	w	2	• Modify file
Execute	x	1	• Run program/script
No permission	-	0	• Denied

Example: chmod 755 script.sh

Owner=7 (rwx) | Group=5 (r-x) | Others=5 (r-x)

Owner=7 (rwx) | Group=5 (r-x) | Others=5 (r-x)

Topic 2 Summary: Key Takeaways

- The Operating System (OS) is the ultimate security gatekeeper. It manages processes, memory, files, devices, and user access.

- Windows is the dominant desktop OS. Critical directories for security include System32, AppData, and Temp.

- Windows Command Prompt commands like ipconfig, netstat, tasklist, systeminfo, and net user are essential for investigation.

- The Principle of Least Privilege means users should have only the minimum access needed. Administrator accounts should be used sparingly.

- Linux dominates servers and cybersecurity tools. Key directories include /etc/ (configuration), /var/log/ (logs), /tmp/ (temporary files), and /home/ (user directories).

- Essential Linux commands include ls -la, cat, grep, ps aux, chmod, whoami, and history.

- Linux file permissions use read (r=4), write (w=2), and execute (x=1) for owner, group, and others. Misconfigured permissions are a common attack vector.